Recover Mission Simulation Report

Route: EGPF EGHH | Distance: 319.3 NM | Bearing: 162.0

Generated: 2026-04-29 23:12

Airports

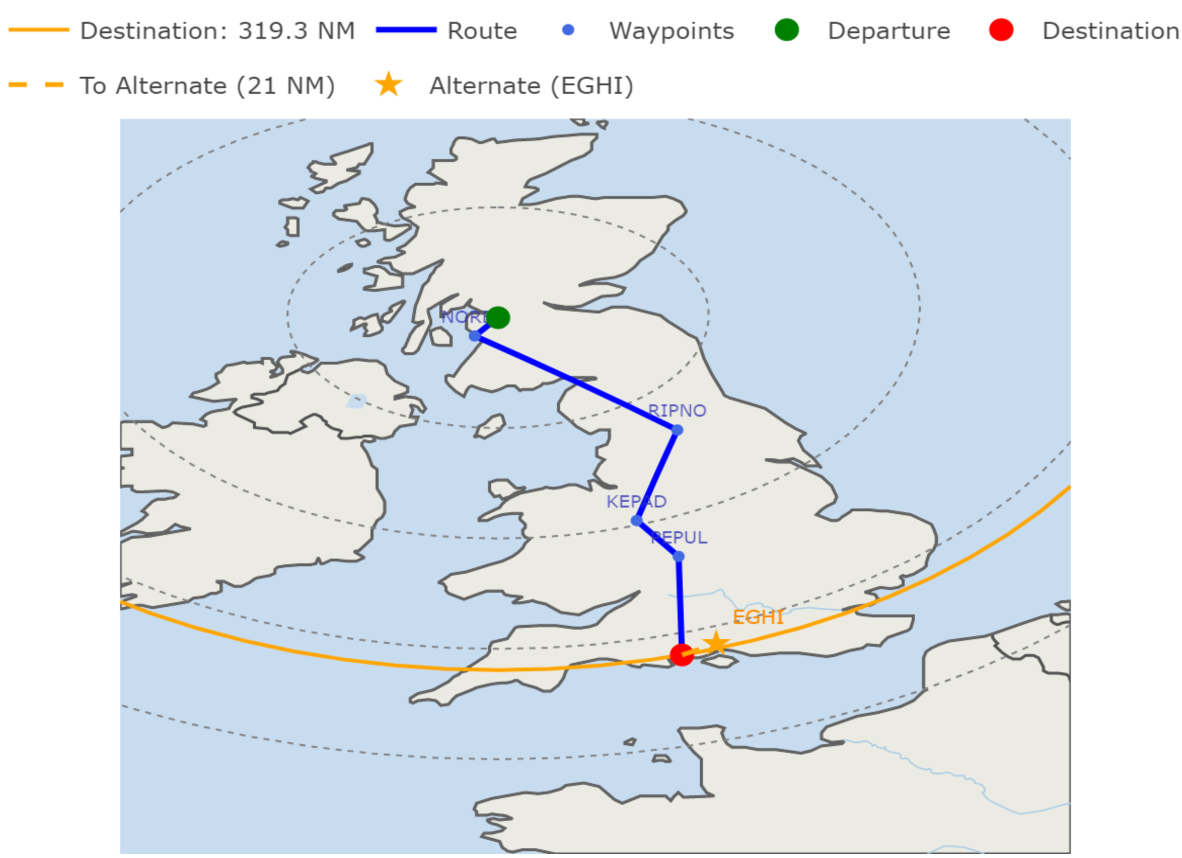
Type	Airport	ICAO	PA (ft)	DA (ft)
Departure	Glasgow Arpt	EGPF	26	26
Arrival	Bournemouth Arpt	EGHH	38	38

Route Distance: 319.3 NM | Bearing: 162.0°

Climb Speed Schedule

Configuration	Best Rate Climb	Cruise Climb
Tamarack M2	105-202 KIAS / M 0.16-0.42	180 KIAS below transition / M 0.51 above

Flight Route Map



Weight Status

Weight Status

Component	Weight (lb)	Max Weight (lb)
BEW	6,972	6,972
Payload	1,350	1,828
Initial Fuel	2,478	3,440
Reserve Fuel	900	600
Taxi Fuel	100	100
Mission Fuel	1,478	1,478
ZFW	8,322	8,800
Ramp Weight	10,800	10,800
Takeoff Weight	10,700	10,700

Tamarack

Takeoff

Climb

Cruise

Descent

Landing

Totals

Flatwing

Takeoff

- Takeoff Start Weight (lb): 10700
- Takeoff End Weight (lb): 10689
- Takeoff Roll Dist (ft): 1106
- Dist to 35 ft (ft): 1320
- Segment 1 Gradient (%): 30.24
- Dist to 400 ft (ft): 3465
- Segment 2 Gradient (%): 29.82
- Dist to 1500 ft (ft): 10468
- Segment 3 Gradient (%): 27.47
- Fuel Remaining After Takeoff (lb): 2367
- Takeoff V-Speeds:
  - Weight: 10689
  - VR: 91.9
  - V1: 91.9
  - V2: 105.0
  - V3: 113.8

Climb

- Climb Time (min): 20
- Climb Dist (NM): 81
- Climb Fuel (lb): 508
- Fuel Remaining After Climb (lb): 1879

Cruise

- Cruise Time (min): 16
- Cruise Dist (NM): 110
- Cruise Fuel (lb): 257
- Cruise VKTAS (knots): 362
- Cruise Max Mach: 0.703
- Time to Target Mach (min): 4.6
- Cruise - First Level-Off Alt (ft): 33000
- Fuel Remaining After Cruise (lb): 2120

Descent

- Descent Time (min): 27
- Descent Dist (NM): 125
- Descent Fuel (lb): 142
- Descent Knee Alt (ft): 30494
- Fuel Remaining After Descent (lb): 2241

Landing

- Landing Start Weight (lb): 9797
- Landing End Weight (lb): 9790
- Landing - Dist from 35 ft to Stop (ft): 1257
- Landing - Ground Roll (ft): 778
- Fuel Remaining After Landing (lb): 1468
- Approach V-Speeds:
  - Weight: 9791
  - VAPP: 119.1
  - VREF: 92.9

Totals

Total Time (min): 64
Simulation Runtime (s): 2.651
Total Dist (NM): 318
Total Fuel Burned (lb): 909
Fuel Remaining (lb): 1468
Integration Steps: 3872
30s Step Count: 20
V1 Cut: False

Hourly Fuel Burn (lb)

Hour	Tamarack (lb)	Flatwing (lb)
1		508/392/900
2		9

Aircraft Configuration: Tamarack M2 (M2 / Tamarack)

Airframe & Aerodynamics

Wing Area: 250.0 ft2
Span: 52.0 ft
AR: 10.82
Oswald e: 0.8025
Sweep 25%c: 0.0 deg
Cd0: 0.02722
CL0: 0.210
CLa: 4.815
CLmax Clean: 1.444
CLmax F15: 1.648
CLmax F40: 1.873

Weights

BEW: 6,972 lb
MZFW: 8,800 lb
Max Payload: 1,828 lb
MTOW: 10,700 lb
MRW: 10,800 lb
Max Fuel: 3,440 lb
Taxi Fuel: 100 lb
Reserve Fuel: 600 lb

Performance Limits

VMO: 263 KCAS
Mmo: 0.71
Ceiling: 41,000 ft
Climb KCAS/Mach: 180 / 0.51
Descent KCAS/Mach: 260 / 0.70
Min ROC: 300 ft/min

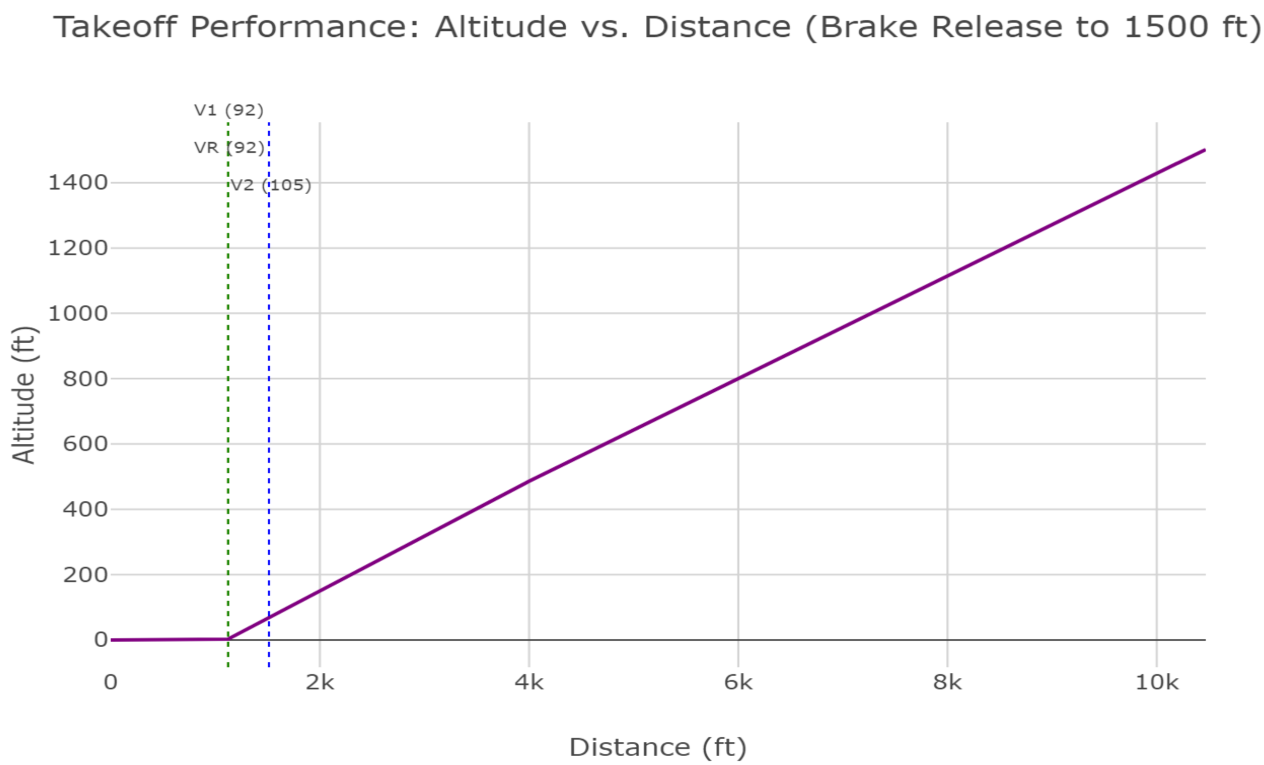
Powerplant

Engine: Williams FJ44-1AP-21
Engines: 2
SFC: 0.6000 lb/hr/lb
Thrust Mult: 0.723
Static Thrust/Eng: 1,965 lb
Total Static Thrust: 3,930 lb

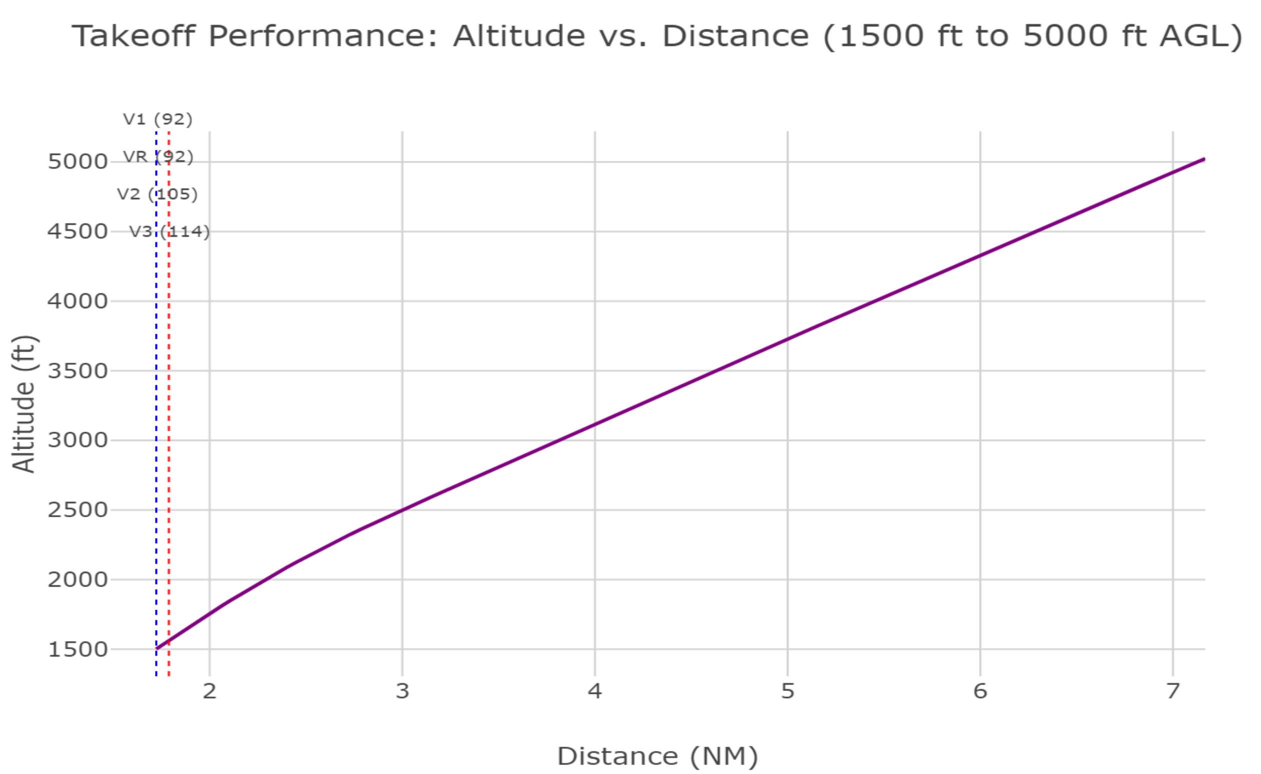
FJ44-class Thrust Deck x 0.723 (lbf/engine)

Alt (ft)	M 0.01	M 0.10	M 0.15	M 0.20	M 0.25	M 0.30	M 0.35	M 0.40	M 0.45	M 0.50	M 0.55	M 0.60	M 0.65
0	2,038	1,872	1,803	1,735	1,662	1,590	1,536	1,482	1,420	1,359	1,337	1,315	1,315
5,000	1,843	1,704	1,647	1,590	1,531	1,472	1,422	1,373	1,333	1,294	1,281	1,268	1,263
10,000	1,648	1,536	1,491	1,446	1,399	1,353	1,309	1,265	1,247	1,229	1,225	1,221	1,211
15,000	1,431	1,370	1,336	1,301	1,258	1,215	1,180	1,145	1,128	1,111	1,101	1,091	1,080
20,000	1,214	1,203	1,180	1,156	1,117	1,077	1,051	1,026	1,010	994	978	961	950
25,000	1,004	996	981	967	942	916	900	885	876	867	863	860	856
30,000	795	788	783	777	766	755	750	744	743	741	750	759	762
35,000	636	636	636	636	636	636	636	636	639	643	647	650	656
40,000	491	491	491	491	491	491	491	491	491	491	498	506	513
45,000	368	368	368	368	368	368	368	368	368	368	368	368	375
50,000	216	220	223	226	231	237	239	242	242	244	242	242	250

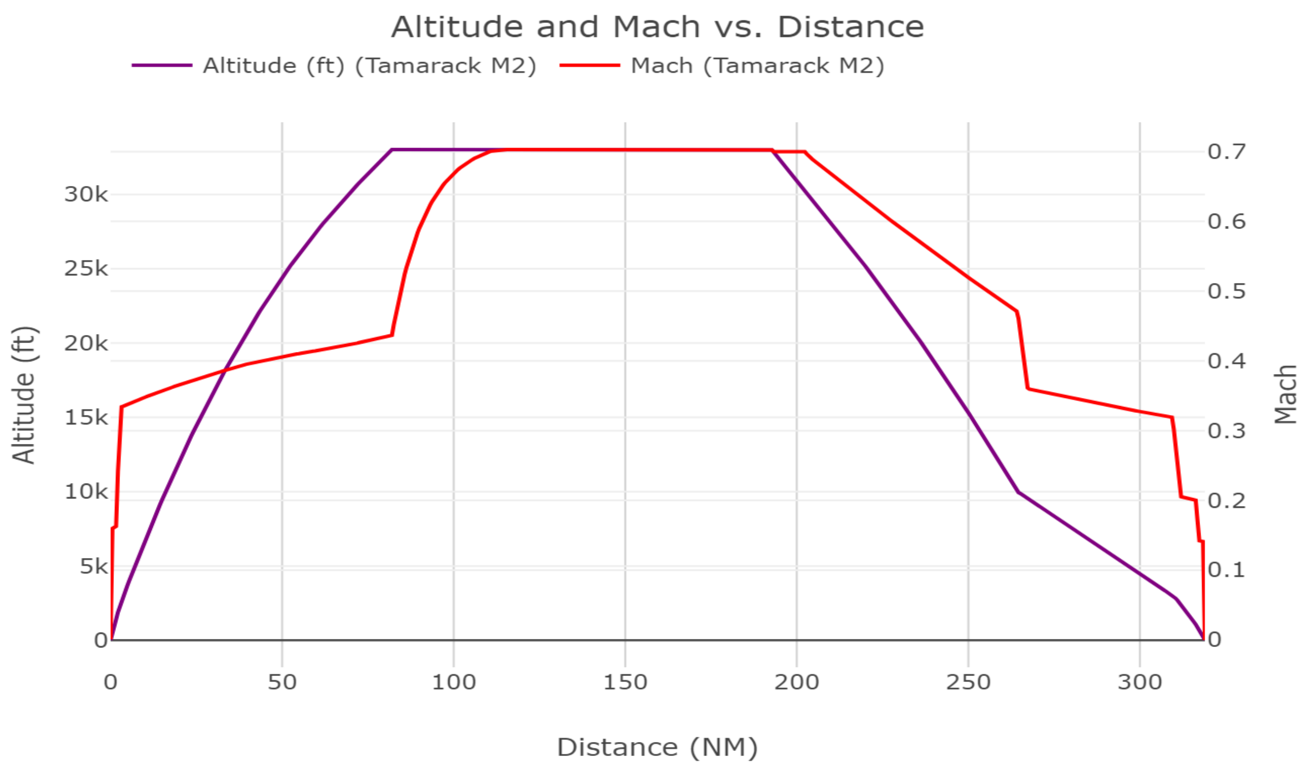
Takeoff Performance: Brake Release to 1500 ft



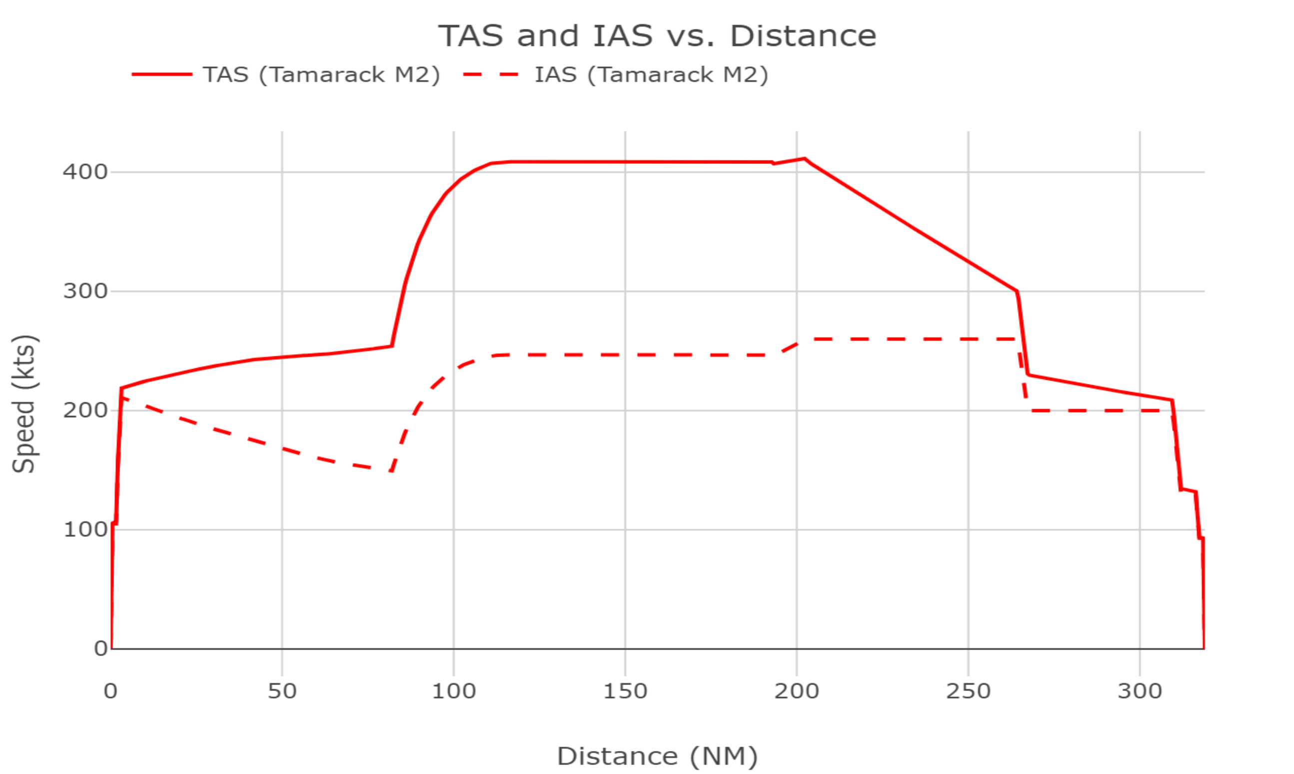
Takeoff Performance: 1500 ft to 5000 ft AGL



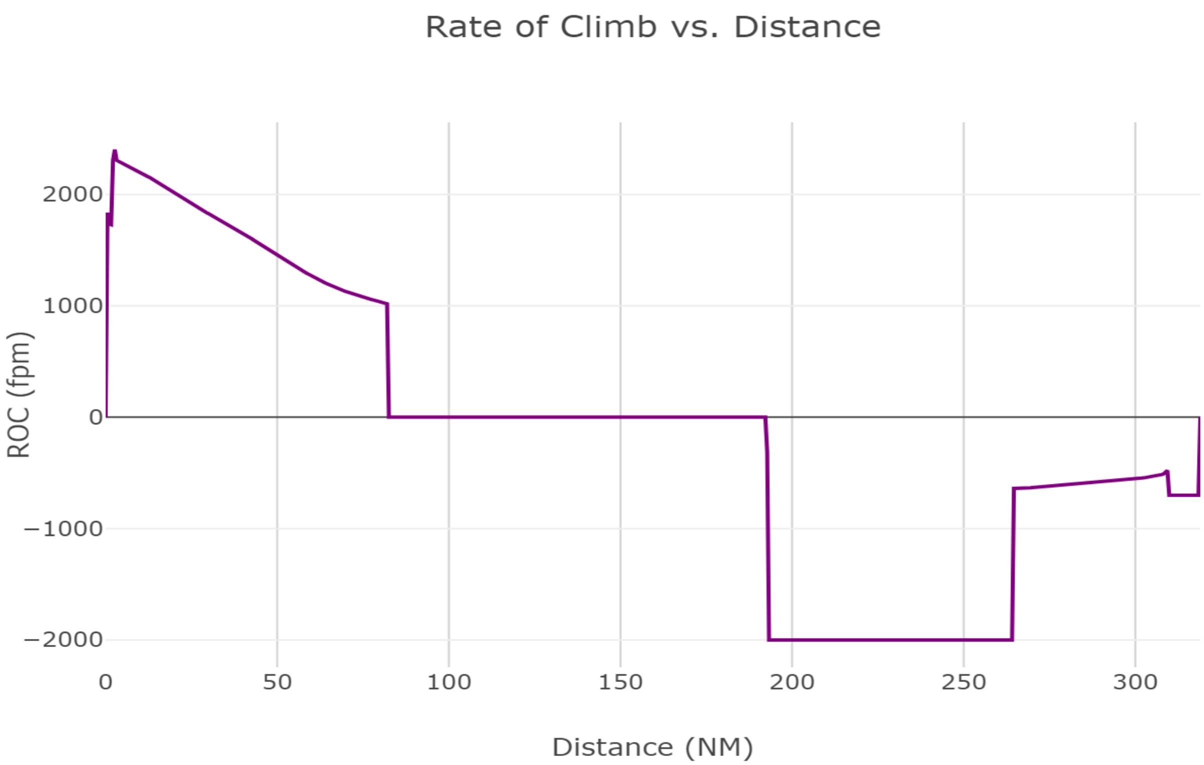
Altitude and Mach vs. Distance



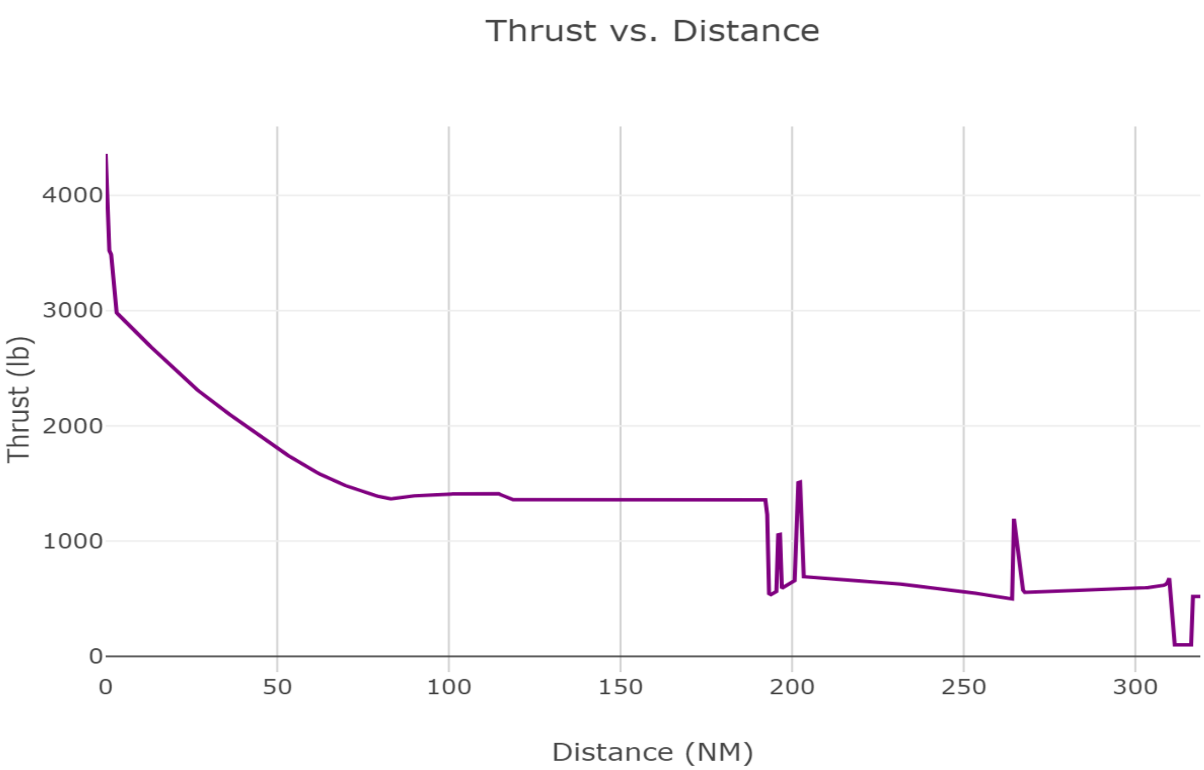
Altitude, TAS and IAS vs. Distance



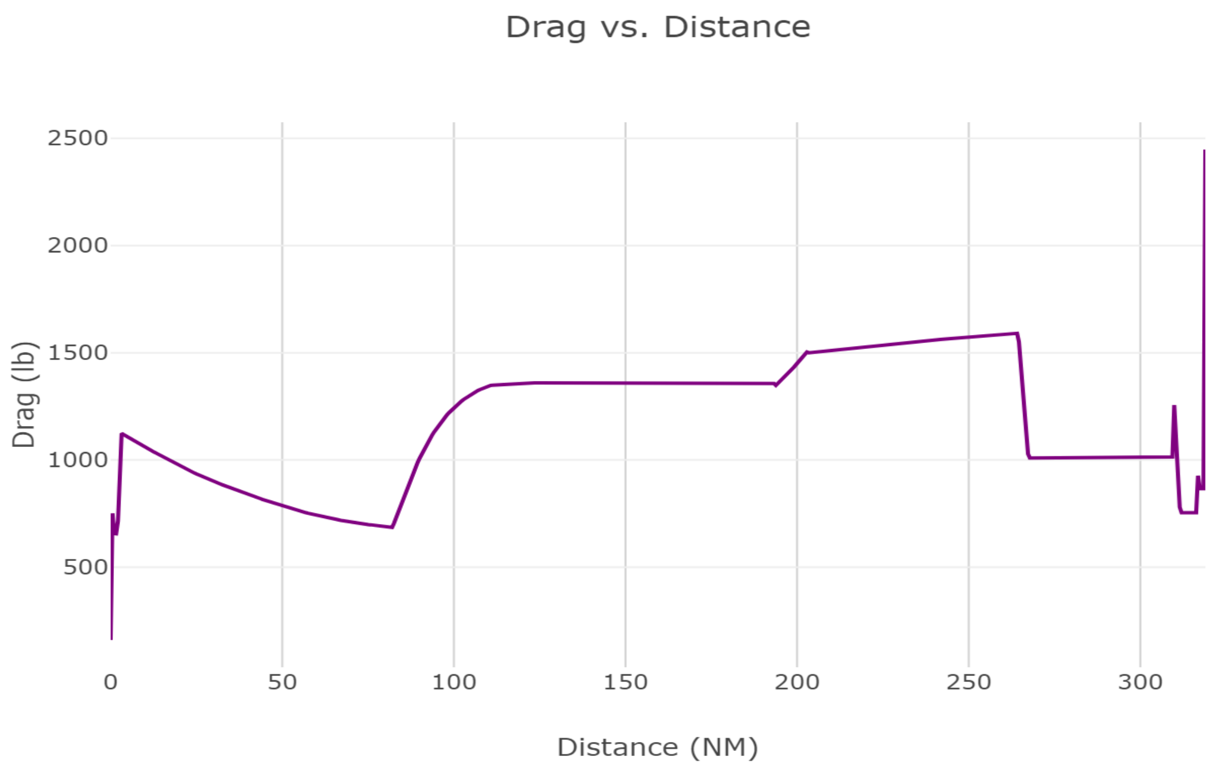
Rate of Climb vs. Distance



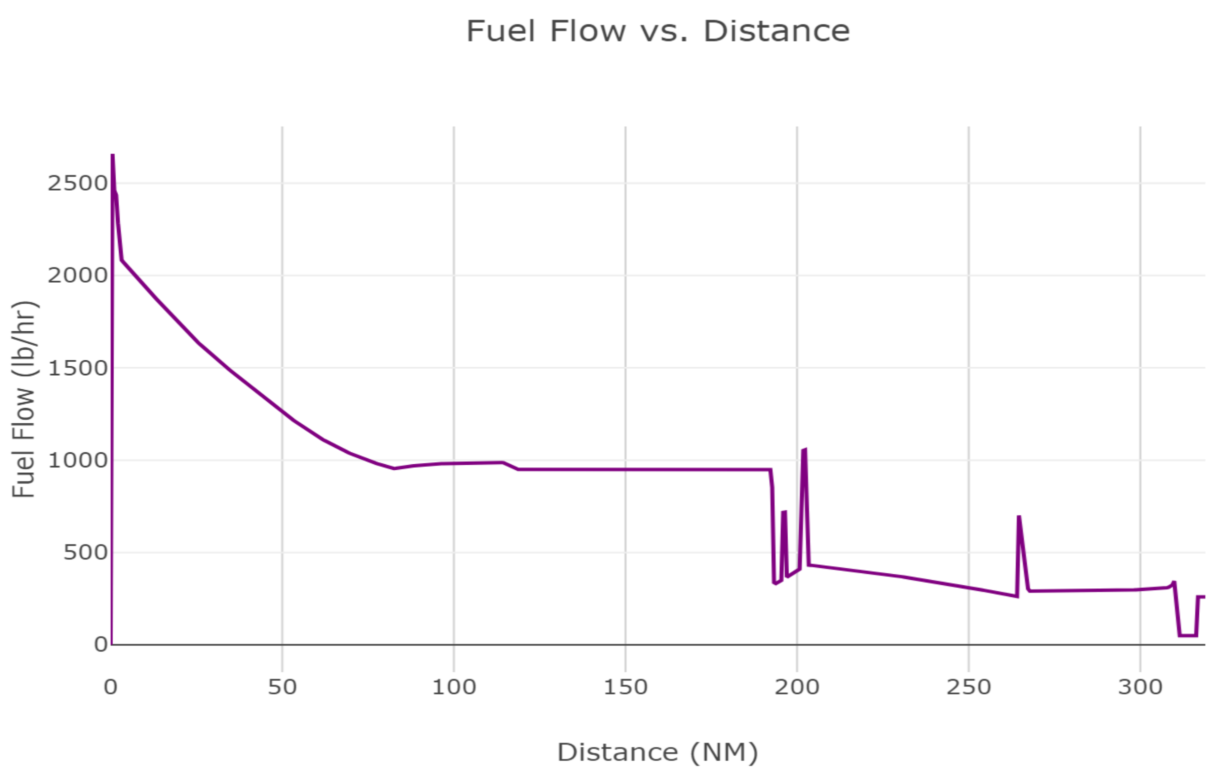
Thrust vs. Distance



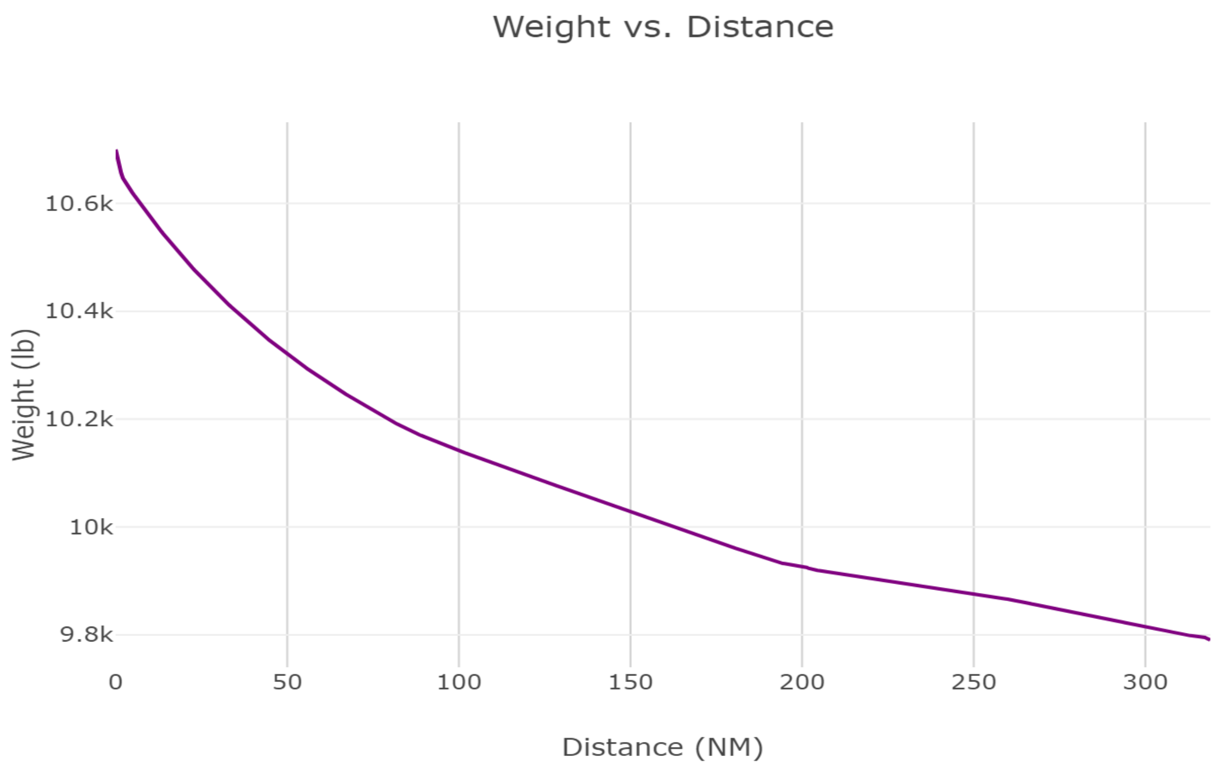
Drag vs. Distance



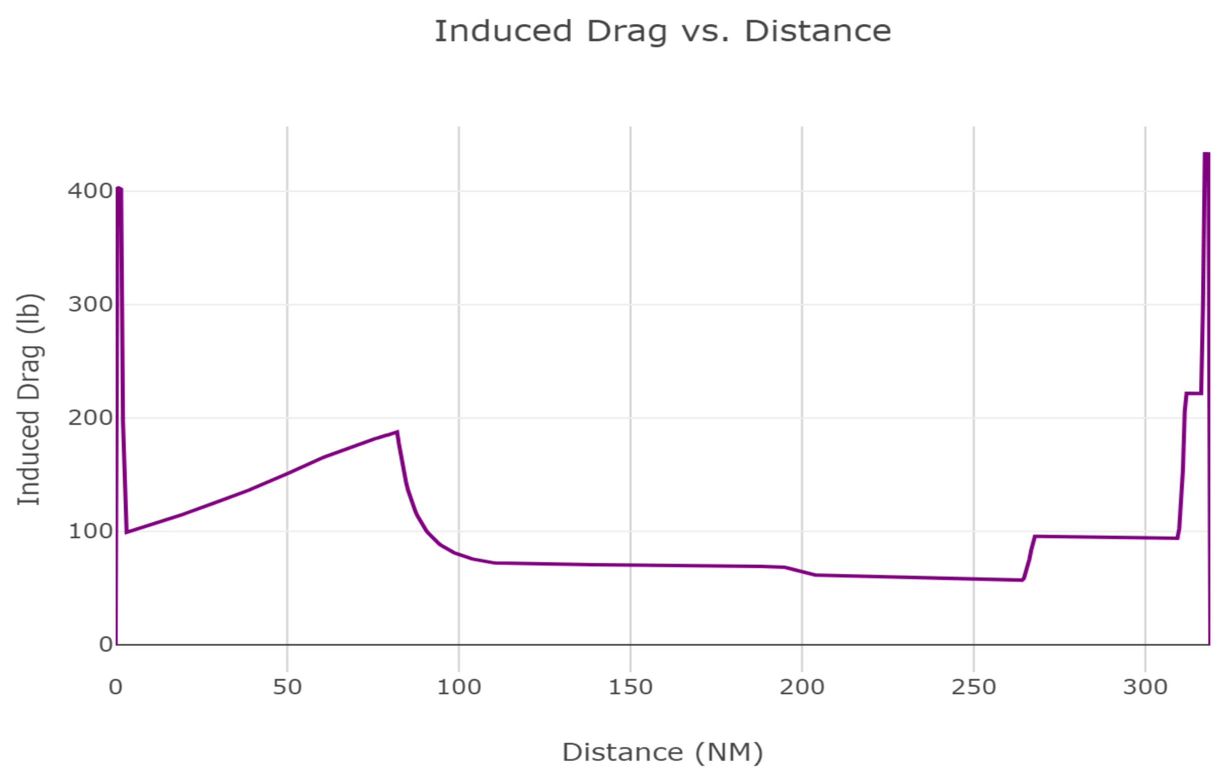
Fuel Flow vs. Distance



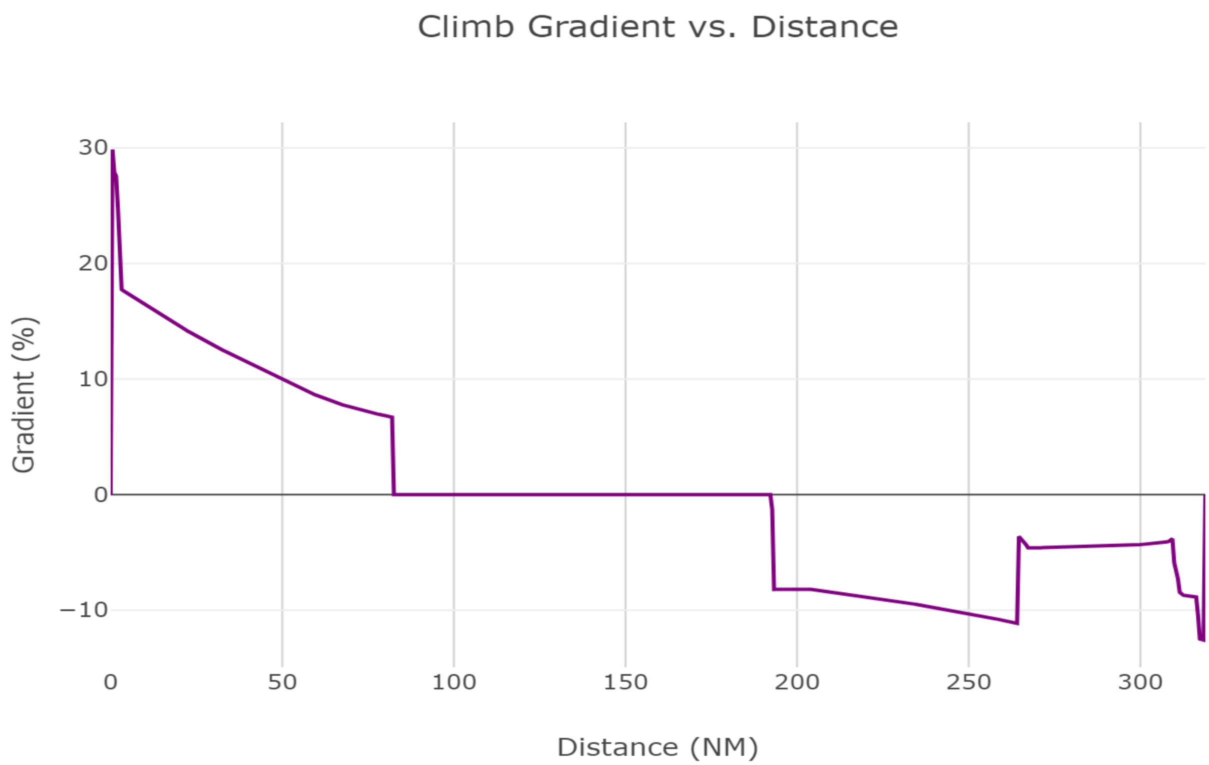
Weight vs. Distance



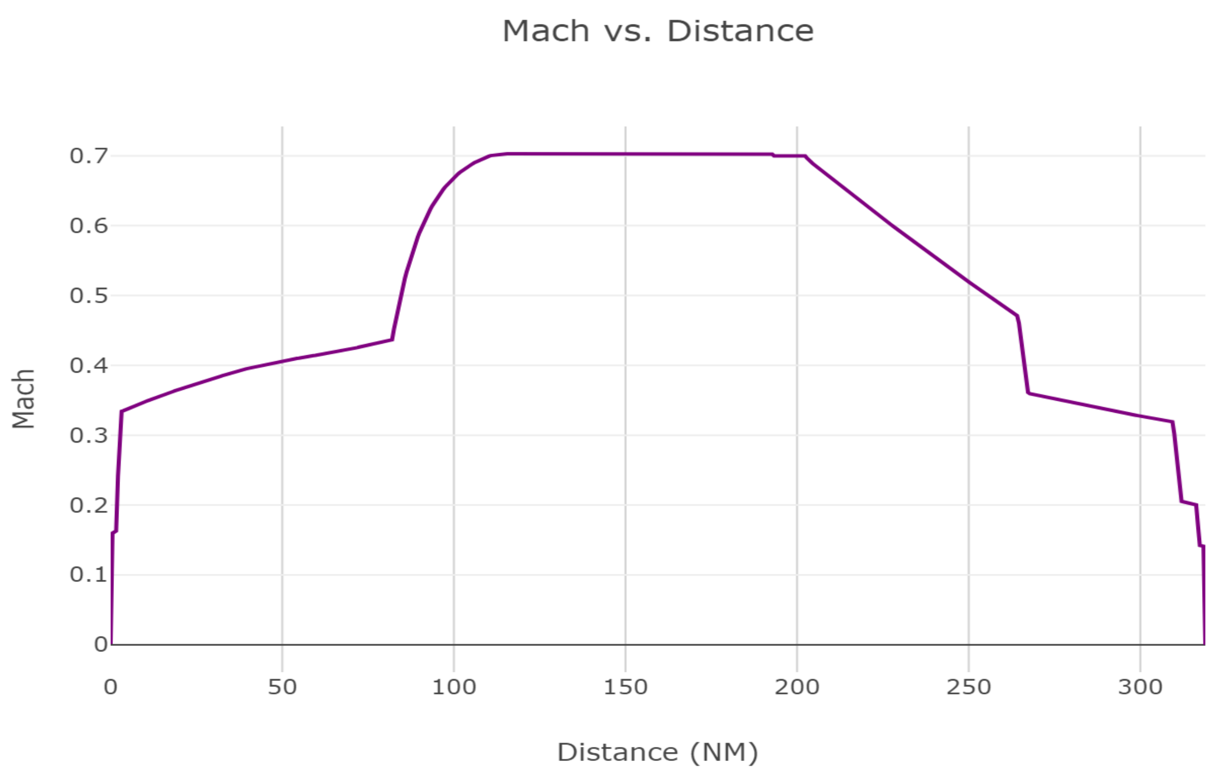
Induced Drag vs. Distance



Climb Gradient vs. Distance



Mach vs. Distance



Fuel Remaining vs. Distance